DDL Commands with Constraints in DBMS (MySQL)

DDL commands create, modify, and delete database structures. Constraints are rules applied to columns to maintain data accuracy and integrity.

The six important SQL constraints are:

Constraint	Purpose
PRIMARY KEY	Uniquely identifies each record; cannot contain NULL.
FOREIGN KEY	Links tables and maintains referential integrity.
UNIQUE	Prevents duplicate non-NULL values.
CHECK	Ensures values satisfy a specified condition.
DEFAULT	Uses a predefined value when no value is provided.
NOT NULL	Prevents NULL values in a column.

1. PRIMARY KEY Constraint

A PRIMARY KEY uniquely identifies each row in a table.

```
CREATE TABLE Students (
StudentID INT PRIMARY KEY,
StudentName VARCHAR(50),
Age INT
);

INSERT INTO Students VALUES (101, 'Arun', 20);
INSERT INTO Students VALUES (102, 'Bala', 21);
INSERT INTO Students VALUES (103, 'Charan', 19);
```

Output:

```
mysql> CREATE TABLE Students (StudentID INT PRIMARY KEY, StudentName VARCHAR(50), Age INT);
Query OK, 0 rows affected
mysql> INSERT INTO Students VALUES (101, 'Arun', 20), (102, 'Bala', 21), (103, 'Charan', 19);
Query OK, 3 rows affected
mysql> SELECT * FROM Students;
+-----------+-------------+-----+
| StudentID | StudentName | Age |
+-----------+-------------+-----+
|       101 | Arun        |  20 |
|       102 | Bala        |  21 |
|       103 | Charan      |  19 |
+-----------+-------------+-----+
3 rows in set
```

Violation:

```
INSERT INTO Students VALUES (101, 'David', 22);
```

```
mysql> INSERT INTO Students VALUES (101, 'David', 22);
ERROR 1062 (23000): Duplicate entry '101' for key 'students.PRIMARY'
```

2. FOREIGN KEY Constraint

A FOREIGN KEY establishes a relationship between two tables.

```
CREATE TABLE Departments (
DepartmentID INT PRIMARY KEY,
DepartmentName VARCHAR(50)
);

CREATE TABLE Employees (
EmployeeID INT PRIMARY KEY,
EmployeeName VARCHAR(50),
DepartmentID INT,
FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID)
);
```

Output:

```
mysql> SELECT * FROM Departments;
+--------------+---------------+
| DepartmentID | DepartmentName |
+--------------+---------------+
|           10 | CSE           |
|           20 | ECE           |
|           30 | IT            |
+--------------+---------------+
3 rows in set
mysql> SELECT * FROM Employees;
+------------+-------------+-------------+
| EmployeeID | EmployeeName | DepartmentID |
+------------+-------------+-------------+
|        101 | Ravi         |          10 |
|        102 | Priya        |          20 |
|        103 | Kumar        |          10 |
+------------+-------------+-------------+
3 rows in set
```

Violation:

```
INSERT INTO Employees VALUES (104, 'Anand', 50);
```

```
mysql> INSERT INTO Employees VALUES (104, 'Anand', 50);
ERROR 1452 (23000): Cannot add or update a child row:
a foreign key constraint fails
```

DepartmentID 50 does not exist in the Departments table.

3. UNIQUE Constraint

The UNIQUE constraint prevents duplicate non-NULL values in a column.

```
CREATE TABLE Users (
UserID INT PRIMARY KEY,
UserName VARCHAR(50),
Email VARCHAR(100) UNIQUE
);

INSERT INTO Users VALUES
(1, 'Arun', 'arun@gmail.com'),
(2, 'Bala', 'bala@gmail.com');
```

Output:

```
mysql> SELECT * FROM Users;
+--------+----------+----------------+
| UserID | UserName | Email          |
+--------+----------+----------------+
|      1 | Arun     | arun@gmail.com  |
|      2 | Bala     | bala@gmail.com  |
+--------+----------+----------------+
2 rows in set
```

Violation:

```
INSERT INTO Users VALUES (3, 'Charan', 'arun@gmail.com');
```

```
mysql> INSERT INTO Users VALUES (3, 'Charan', 'arun@gmail.com');
ERROR 1062 (23000): Duplicate entry 'arun@gmail.com' for key 'users.Email'
```

4. CHECK Constraint

The CHECK constraint ensures that data satisfies a particular condition. Students' ages must be at least 18.

```
CREATE TABLE CollegeStudents (
StudentID INT PRIMARY KEY,
StudentName VARCHAR(50),
Age INT CHECK (Age >= 18)
);

INSERT INTO CollegeStudents VALUES (101, 'Arun', 20);
```

Valid insertion output:

```
mysql> INSERT INTO CollegeStudents VALUES (101, 'Arun', 20);
Query OK, 1 row affected
```

Invalid insertion:

```
INSERT INTO CollegeStudents VALUES (102, 'Bala', 16);
```

```
mysql> INSERT INTO CollegeStudents VALUES (102, 'Bala', 16);
ERROR 3819 (HY000): Check constraint is violated.
```

Another example: Marks INT CHECK (Marks BETWEEN 0 AND 100) allows marks from 0 to 100. Add NOT NULL if a value is mandatory.

5. DEFAULT Constraint

The DEFAULT constraint assigns a predefined value when a column is omitted from an INSERT statement.

```
CREATE TABLE Orders (
OrderID INT PRIMARY KEY,
ProductName VARCHAR(50),
Status VARCHAR(20) DEFAULT 'Pending'
);

INSERT INTO Orders (OrderID, ProductName)
VALUES (101, 'Laptop');

INSERT INTO Orders VALUES (102, 'Mouse', 'Delivered');
```

Output:

```
mysql> SELECT * FROM Orders;
+---------+-------------+-----------+
| OrderID | ProductName | Status    |
+---------+-------------+-----------+
|     101 | Laptop      | Pending   |
|     102 | Mouse       | Delivered |
+---------+-------------+-----------+
2 rows in set
```

For OrderID 101, Status was omitted, so MySQL used Pending. For OrderID 102, Delivered was explicitly provided.

6. NOT NULL Constraint

The NOT NULL constraint ensures that a column cannot contain NULL values.

```
CREATE TABLE Teachers (
TeacherID INT PRIMARY KEY,
TeacherName VARCHAR(50) NOT NULL,
Salary DECIMAL(10,2)
);

INSERT INTO Teachers VALUES (101, 'Ravi', 45000);
```

Valid insertion output:

```
mysql> INSERT INTO Teachers VALUES (101, 'Ravi', 45000);
Query OK, 1 row affected
```

Invalid insertion:

```
INSERT INTO Teachers VALUES (102, NULL, 50000);
```

```
mysql> INSERT INTO Teachers VALUES (102, NULL, 50000);
ERROR 1048 (23000): Column 'TeacherName' cannot be null
```

Adding Constraints Using ALTER TABLE

Constraints can also be added after creating a table.

```
ALTER TABLE Students ADD PRIMARY KEY (StudentID);

ALTER TABLE Students ADD CONSTRAINT fk_dept FOREIGN KEY (DeptID)
REFERENCES Departments(DeptID);

ALTER TABLE Students ADD UNIQUE (Email);

ALTER TABLE Students ADD CONSTRAINT chk_age CHECK (Age >= 18);

ALTER TABLE Students ALTER COLUMN City SET DEFAULT 'Chennai';

ALTER TABLE Students MODIFY StudentName VARCHAR(50) NOT NULL;
```

Terminal-style output:

```
mysql> ALTER TABLE Students ADD UNIQUE (Email);
Query OK, 0 rows affected
mysql> ALTER TABLE Students ADD CONSTRAINT chk_age CHECK (Age >= 18);
Query OK, 0 rows affected
mysql> ALTER TABLE Students MODIFY StudentName VARCHAR(50) NOT NULL;
Query OK, 0 rows affected
```

Note: The ALTER TABLE commands require the referenced columns to exist and the data to satisfy the constraints.